

Variational framework for infinitely deep transformers in the mean-field regime of tokens

Goal

Propose a framework to study gradient descent algorithm for infinitely deep transformers in the mean-field regime of tokens.

The model

- **Infinitely deep transformers** : Transformers with infinitely many layers. The tokens evolve through the transformer as a solution to an ODE.

$$\dot{x}_t = v_t(x_t, m_t).$$

- **Mean-field regime** : The attention mechanism depends on a probability measure. This probability measure evolves through the transformer according to a continuity equation.

$$\partial_t m_t + \nabla v_t(\cdot, m_t) \cdot m_t = 0.$$

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Parameterization

Parameters of our network : $v \in E := L^2([0, 1], V)$ where V is an operator-valued RKHS.

It represents the function guiding the two equations presented in the last slide. See it as the composition of an attention layer and a MLP layer.

The risk

For $v \in E$, the risk writes

$$R(v) = \mathbb{E}_{(\bar{x}, \bar{m})} \left[l \left(\phi_{0,1}^{v, \bar{m}}(\bar{x}), \bar{m}, \bar{x} \right) \right].$$

Where $(\bar{x}, \bar{m})$ is a random variable satisfying $\bar{x} \sim \bar{m}$, l is any loss function and $\phi_{0,1}^{v, \bar{m}}(\bar{x})$ is the output of the transformer for initial token $\bar{x}$ and initial measure $\bar{m}$.

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The gradient

Under regularity assumptions on V and I , the risk is differentiable.

Stochastic Gradient Descent (SGD) for linear attention

- **Linear attention** : Consider the RKHS

$$V = \left\{ (x, y) \mapsto \frac{1}{H} \sum_{h=1}^H \langle Ax, y \rangle W y, H \in \mathbb{N}^*, A \in \mathcal{M}_d(\mathbb{R}), W \in \mathcal{M}_d(\mathbb{R}) \right\}.$$

- Encoding the matrices A and W in P and S , we propose the algorithm

$$\begin{aligned} \Upsilon_t^k &= (1 - \gamma_k \eta) P_t^k S_t^{kT} - \gamma_k \bar{p}_t^k \bar{s}_t^{kT}, \\ P_t^{k+1} &= P_t^k + \Delta P_t^k, S_t^{k+1} = S_t^k + \Delta S_t^k, \\ \begin{cases} \Delta P_t^k &= -\gamma(\eta P_t^k S_t^{kT} + \bar{p}_t^k \bar{s}_t^{kT}) S_t^k (S_t^{kT} S_t^k + \lambda_P I_H)^{-1}, \\ \Delta S_t^k &= [-\gamma(\eta P_t^k S_t^{kT} + \bar{p}_t^k \bar{s}_t^{kT}) - \Delta P_t^k S_t^{kT}]^T P_t^k (P_t^{kT} P_t^k + \lambda_S I_H)^{-1} \end{cases} \end{aligned}$$